

Muhua (黄牧华) Huang

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Computational social scientist and Stanford Organizational Behavior PhD researcher studying how AI systems represent human traits and reshape collective behavior. I build large-scale LLM-agent experiments and psychometric methods to investigate human-AI collaboration, culture, values, and emergent social dynamics.

EDUCATION

Stanford Graduate School of Business, PhD in Organizational Behavior	2025 – 2030
University of Chicago, MA in Computational Social Science, GPA: 3.9/4.0	2023 – 2025
University of British Columbia, BA in Computer Science & Psychology (Honours), GPA: 4.2/4.3	2018 – 2023

Additional training: International AI Evaluation Programme Fellow; SICSS-Beijing (Outstanding Project, scholarship); Google Research Mentorship Program Scholar.

INDUSTRY EXPERIENCE

Student Researcher Google DeepMind	Jun – Sep 2026
Contributing Researcher and Writer Stanford HAI & Google DeepMind	2026
Co-developed <i>Organizing Intelligence</i>, contributing to conceptualization, research, and writing of a 16-topic AI and organizations primer.	
Research Intern Microsoft Research Asia	Jul – Oct 2024
- Built a scalable multi-agent platform with cognitive modules, RAG, and parallel processing to study how LLM communities develop values and social structures. - Designed experiments varying value diversity, population size, and time; analyzed network change and emergent language, culture, and institutions with social network analysis, GraphRAG, and NLP.	

RESEARCH EXPERIENCE

PhD Student Researcher Stanford Graduate School of Business	2025 – Present
Ran approximately 375,000 GPT-4o trials across networks of 2–50 agents; identified human-AI differences in category boundaries and a 12.7% performance drop when agents were constrained to human labels.	
Master's Researcher University of Chicago	2023 – 2025
Developed a validated Big Five framework for LLM agents; data from 350 participants showed personality predicts agents' moral reasoning, risk-taking, and cooperation.	
Research Assistant University of British Columbia	2020 – 2023
Applied SEM, simulation, ML/NLP, and eye-tracking across personality measurement, youth essay analysis (10,000+ essays; supervised 20 RAs), and personalized AI explanations; work contributed to three journals and an IJCAI workshop.	

SELECTED PUBLICATIONS

AI Simulation: Huang, M., & Evans, J. (2026). Institutions as cached computation for resource-rational negotiation. *Behavioral and Brain Sciences*.

AI Character/Value Alignment: Huang, M., et al. (2026). Designing AI-Agents with Personalities. *Personality Science*; Bai, Y., Duan, S., Huang, M., et al. (2026). IROTE. *AAAI*.

Social Science: Savalei, V., & Huang, M. (2025). Fit Indices Are Insensitive to Multiple Minor Violations of Perfect Simple Structure. *Psychological Methods*.

LEADERSHIP & SERVICE

Computational Social Science Workshop Coordinator University of Chicago	2024 – 2025
Organized the weekly MACSS workshop featuring speakers on advanced computational methods and interdisciplinary research.	
Teaching Assistant University of British Columbia	2021 – 2022
Taught Human Computer Interaction Methods and Systematic Program Design; rated 4.7/5 and received three return offers.	
Crisis Respondent Kids Help Phone; 230+ hours and 550+ people supported	2019 – 2020
Peer Reviewer <i>Personality Science</i> ; <i>Behaviour & Information Technology</i>	
Methods: LLM agents/GABM, psychometrics/SEM, social network analysis, ML/NLP, behavioral experiments, RAG, Python, R, SQL.	
Recognition: Stanford GSB & EDGE Fellowships; UChicago Thesis Award; MSRA Outstanding Intern; OpenAI Researcher Access.	